林口長庚 復健科
Orientation



病房CR 鄭德葳 19172

行程

病房每日行程表

	W1	W2	W3	W4	W5
AM			**07:30** **晨會**		
PM	**12:10** **跨團隊會議** 完成結構化病歷(clerk)	跨團隊會議單(PGY) 下班前交給CR			**12:10** **學術會議**

- 上班時間：07:30-17:00 (**New pt 16:30**)
- 學術會議：Chart round、EBM、吞嚥會議、MM
- 每月一次治療師教學
- 教學門診：**務必準時出席，勿跑錯地方****

8月會議、教學安排

日期	時間	活動	報告者
QW1	**12:10**	跨團隊會議	**All PGY&clerk**
09/02	**12:10**	治療師教學	
09/04	**12:10**	**Chart run**	**Y2** 鍾哲瑄
09/11	**12:10**	**EBM**	**Y2** 王彥喬
09/18	**12:10**	全人討論會	**Y2** 張欣雅
09/29	**13:00**	**PGY**座談會	

病歴、作業

基本要求（For clerk）

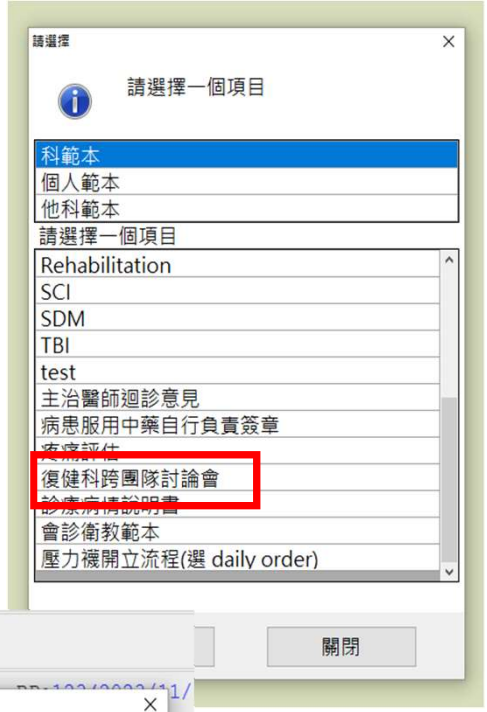
- **出席教學門診**
- 準時出席晨會、學術會議，協助完成會議紀錄
- Primary care 1-2床，病歷書寫
- 電子、紙本護照
- 檢查診（心肺、肌電圖，**需聯絡***）
- 一次CBL（**需聯絡***）

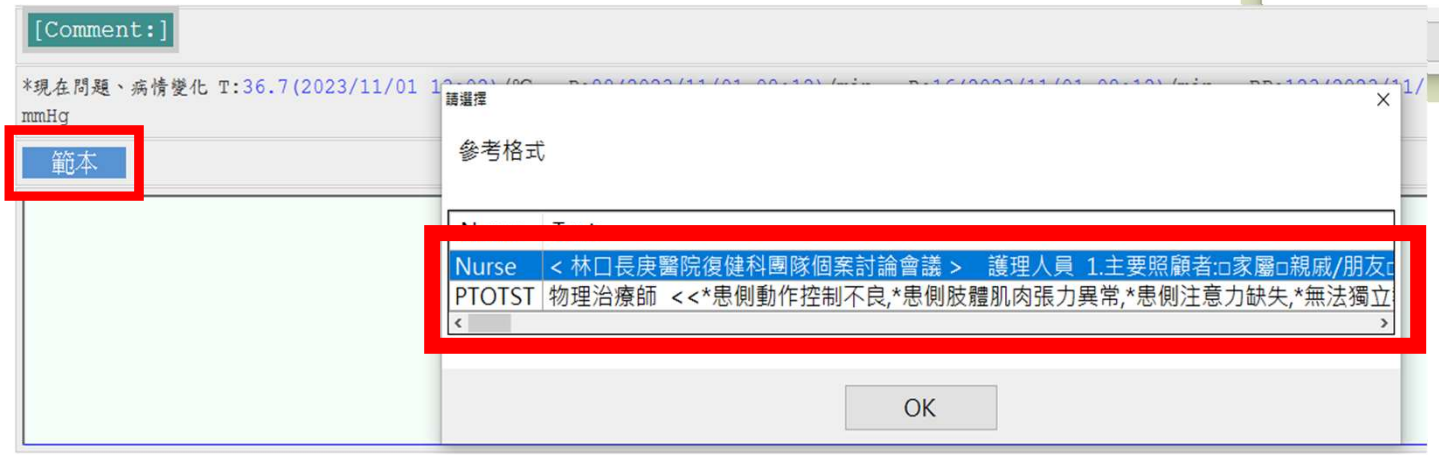
離站時繳交作業（For clerk）

- 紙本學習護照*1
- 教學住診**記錄表***1
- 教學住診**病歷表***1
- Teaching round 紀錄表*1
- 滿意度問卷調查表/實習生回饋表*1
- 病歷迴診教學紀錄表*1(兩個case都要寫)
- CBL評分表(一組1張)

跨團隊會議記錄（For clerk）

- W1會議後，完成自己team討論個案的On duty note
- 路徑: On duty note→科範本→結構化病歷→復健科→跨團隊會議→按照老師、護理師的團隊紀錄勾選→帶入存檔





病歷：Objective 每日需更新不同內容

W1	W2	W3	W4	W5	W6	W7
復健評估	PE	Barthel index	復健紀錄 (PT/OT/ST)	復健評估 +PE+NE	改 S 復健評估	由本科 R/VS 完成
VS comment	Teaching	VS comment	Teaching	VS comment		

- **W6值班PGY醫師請根據當日班別，協助完成以下分層 Progress Note ：**
  - **若L2班：5G**
  - **若L1/EM班：5G + *6H (新)***
- 請記得統一將 S 改為：Activities of daily living impairment

W6
值班Note

職	PGY	PGY	PGY	PGY	PGY	PGY	PGY
GS	18752	15578	18190	23150			
日期	PGY2 鍾	PGY2 李	PGY2 江	PGY2 簡	PGY1 柯	PGY1 陳	PGY1 許
	瑄	嬨	維	山	生	蓉	峻
1	L2						
2				L/EM			
3		L/EM					
4			L2				
5	L2						
6				L2			
7					L2		
8		L/EM					
9	L2						
10						L2	
11					L2		
12							L2
13				L2			
14						L2	
15							L2
16		L2					

L2班：
5G

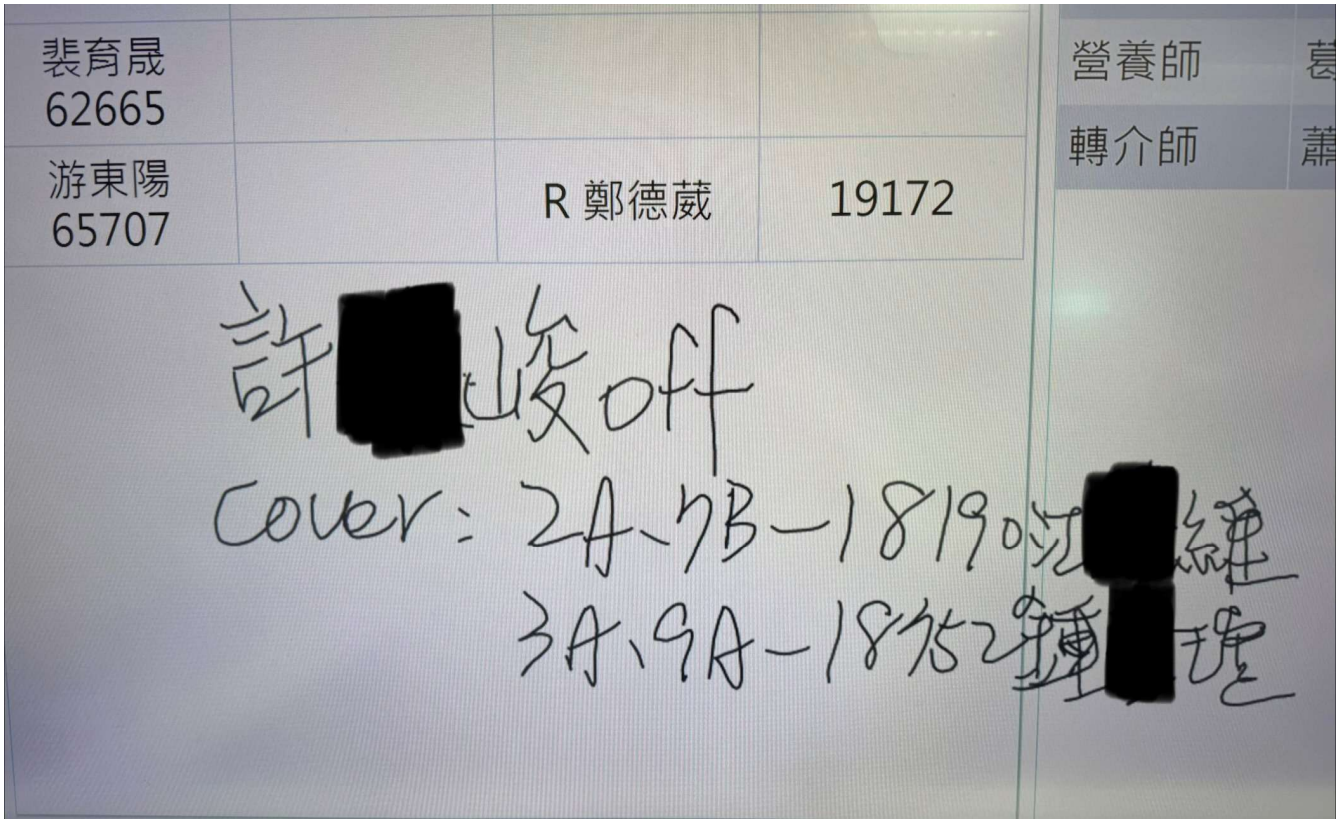
L/EM班：
5G + 6H*

L2班：
5G

好同事，周末不留事

- Foley, NG 到期
- 藥物到期

AM off 麻煩電子白板寫cover team



Assessment 寫法參考

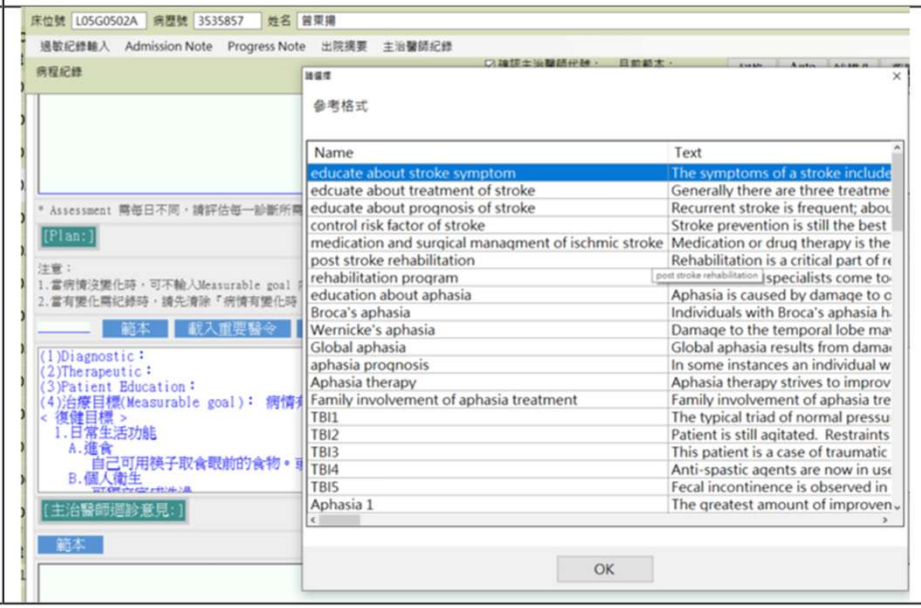
- 盡量不要寫縮寫（可以括號附註）
- 一個完整的復健科診斷包含：臨床診斷, **Onset, 經過哪些處置**, 功能性診斷, 評估量表, **目前的後遺症, mRS**
  - 腦中風：Right thalamic hemorrhage, **onset on 2019/01/02, with left hemiparesis, modified Rankin scale: 3**
  - 創傷性腦傷：Traumatic brain injury with right fronto-temporo-parietal subdural hemorrhage, onset on 2018/11/21, status post craniectomy on 2018/11/21, status post cranioplasty on 2019/02/19, **Rancho Los Amigos level: IV**, with left hemiplegia, dysphagia, modified Rankin scale: 4

Assessment 寫法參考

- 脊髓損傷：Spinal cord injury, onset 2019/06/01, with C3-5 herniated intervertebral disc, C5-6 ruptured disc, spinal compression and myelopathy, status post C3-6 anterior discectomy and interbody fusion with PEEK cage on 2019/06/01, **Neurologic level of injury: C6, American Spinal Injury Association impairment scale: C**, with the sequelae of tetraplegia, mRS:4

Measurable goal & Teaching

- Measurable goal：使用「科範本」內的復健目標：病歷輸入 → progress note → 點選『科範本+Rehabilitation 』→ Plan 點範本 → Measurable Goal (有下拉式選單)
- 主治醫師迴診意見：範本會跳出各式各樣 Teaching

Measurable Goal	主治醫師迴診意見
	

電腦桌面有範本可使用

復健科教學資料

- 住院醫師PRN MSK超音波教學、主題式教學
- 請留意Team+群組通知

學什麼

復健科

各科的好朋友

從少到老
長期陪伴
團隊分工
合作無間



復健科三大疾病

- Stroke (Ischemic/Hemorrhagic)
- Traumatic Brain Injury
- Spinal Cord Injury

- Function
- 全人照護

Rehabilitation assessment

<Rehabilitation assessment>
*Consciousness: mildly stupor, Glasgow coma scale: E4VaM6
*O2: room air
*Intubation: (+) NG, (-) Foley cath (remove today), (-) tracheostomy
*Swallowing: dysphagia, drooling(-), choking(+)
*Speech: aphasia and dysarthria
 naming: poor, 筆(O), 枕頭x
 comprehension: fair, poor , equivocal 1-step order
 fluency: poor, 1-10: x
 repetition: poor-fair, 5 words

*Brunnstrom stage:	R	L	Muscle power:	R	L
UE(P):	I	NP	UE(P):	0	5
UE(D):	I	NP	UE(D):	0	5
LE:	I	NP	LE(P):	0	5
			LE(D):	0	5

*Modified Ashworth Scale	R	L
upper limb:	0	0
lower limb:	0	0

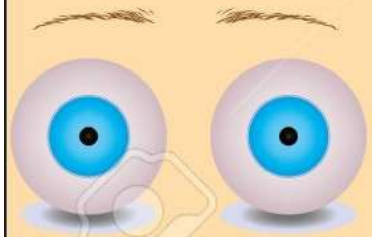
*Urine / stool: incontinence / continence
*Functional status:(unable, poor, fair, good)
 rolling: R2L poor, sitting: poor, standing: unable, walking: unable
*Activities of daily life: totally dependent

Level of consciousness

Awake
Confused
Disoriented
Lethargic
Obtundation
Stuporous
Comatose

Level of consciousness	Description
Awake	Alert, responds immediately and fully to commands — may or may not be fully oriented
Confused	The inability to think rapidly and clearly. There is impaired judgment and decision making
Disoriented	This is the beginning of loss of consciousness. There is disorientation in place, impaired memory and a loss of recognition of self which is the last to deteriorate
Lethargic	Drowsy, sleeps a lot, but is easily aroused with minimal stimuli, i.e. voice, and then responds, but may not be oriented in time, place or person
Obtundation	Can be aroused by stimuli (not pain), i.e. shaking, and will then respond to questions or commands. Remains aroused as long as stimulation is applied, if not will fall asleep, questions are answered with minimal response. During the arousal, patient responds but may be confused
Stuporous	This is a condition of deep sleep or unresponsiveness. The patient can only be aroused or caused to make a motor or verbal response by vigorous and repeated extemal stimulation (painful). The response initiated is often withdrawal or grabbing at stimulus
Comatose	There is no motor response to the external environment or to any stimuli, even deep pain or suctioning. There is no arousal to any stimulus. Reflexes may be present, abnormal movement (posturing) to pain may be present

Glasgow Coma Scale

Behaviour	Response
 Eye Opening Response	4. Spontaneously 3. To speech 2. To pain 1. No response
 Verbal Response	5. Oriented to time, person and place 4. Confused 3. Inappropriate words 2. Incomprehensible sounds 1. No response
 Motor Response	6. Obeys command 5. Moves to localised pain 4. Flex to withdraw from pain 3. Abnormal flexion 2. Abnormal extension 1. No response

Total Score

Best score - 15
Comatosed - ⩽8
Unresponsive - 3

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 fluency: poor, 1-10: x
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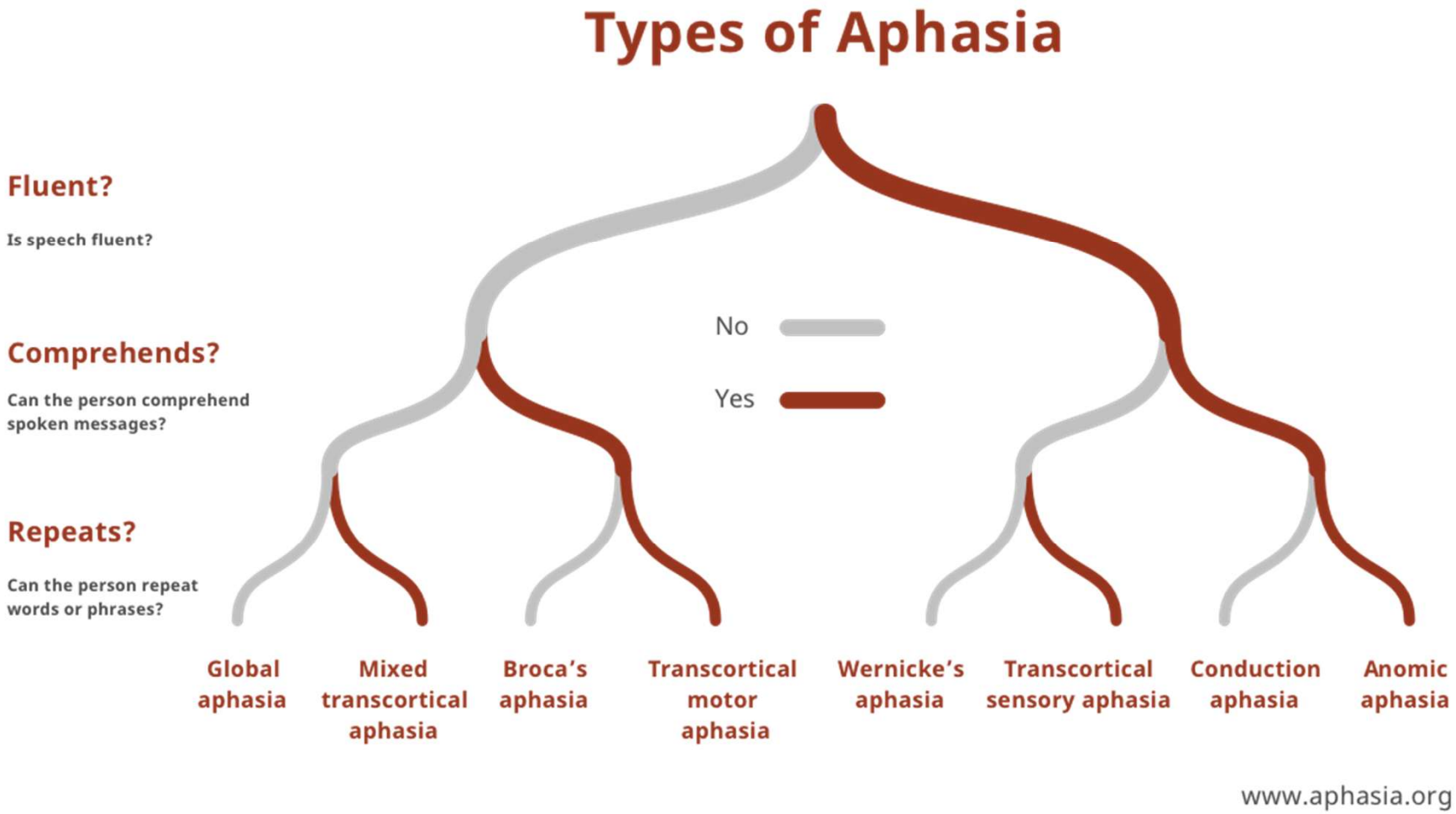
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UE(D):	I	NP	UE(D):	0	5
LE:	I	NP	LE(P):	0	5
			LE(D):	0	5

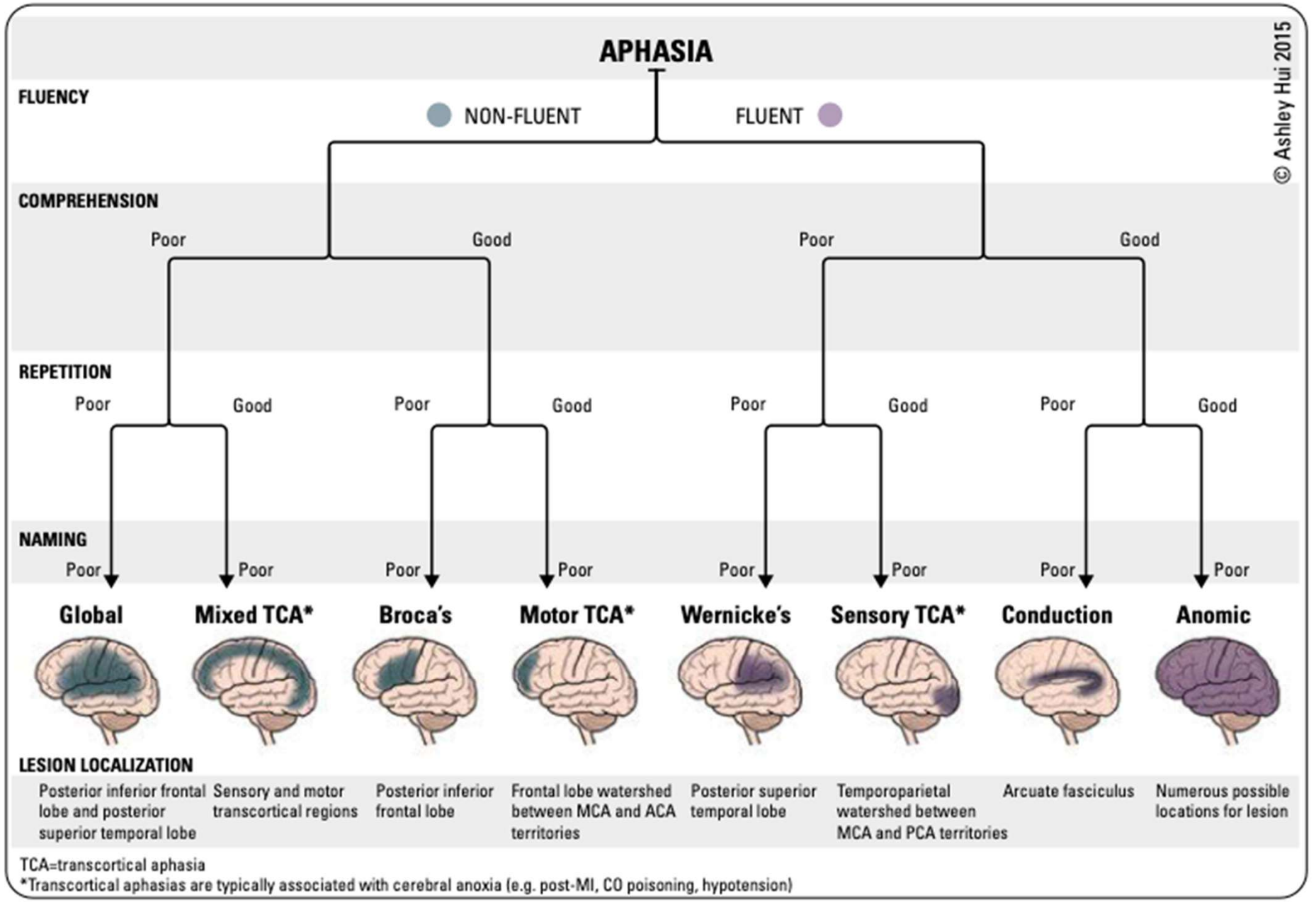
*Modified Ashworth Scale	R	L
upper limb:	0	0
lower limb:	0	0

*Urine / stool: incontinence / continence
*Functional status:(unable, poor, fair, good)
 rolling: R2L poor, sitting: poor, standing: unable, walking: unable
*Activities of daily life: totally dependent

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LE: I NP LE(P): 0 5
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upper limb: 0 0
lower limb: 0 0
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Naming (X)


Types of Aphasia
Fluent?
Is speech fluent?
Comprehends?
Can the person comprehend spoken messages?
Repeats?
Can the person repeat words or phrases?
No
Yes
Global aphasia
Mixed transcortical aphasia
Broca's aphasia
Transcortical motor aphasia
Wernicke's aphasia
Transcortical sensory aphasia
Conduction aphasia
Anomic aphasia
www.aphasia.org


APHASIA
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FLUENCY
NON-FLUENT
FLUENT
COMPREHENSION
Poor
Good
Poor
Good
REPETITION
Poor
Good
Poor
Good
Poor
Good
Poor
Good
NAMING
Poor
Poor
Poor
Poor
Poor
Poor
Poor
Poor
Global
Mixed TCA*
Broca's
Motor TCA*
Wernicke's
Sensory TCA*
Conduction
Anomic
LESION LOCALIZATION
Posterior inferior frontal lobe and posterior superior temporal lobe
Sensory and motor transcortical regions
Posterior inferior frontal lobe
Frontal lobe watershed between MCA and ACA territories
Posterior superior temporal lobe
Temporoparietal watershed between MCA and PCA territories
Arcuate fasciculus
Numerous possible locations for lesion
TCA=transcortical aphasia
*Transcortical aphasias are typically associated with cerebral anoxia (e.g. post-MI, CO poisoning, hypotension)

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UE(P): I NP UE(P): 0 5
UE(D): I NP UE(D): 0 5
LE: I NP LE(P): 0 5
LE(D): 0 5
*Modified Ashworth Scale R L
upper limb: 0 0
lower limb: 0 0
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*Activities of daily life: totally dependent

Swallowing screening

Name:_____
Date:_____
Time:_____

GUSS

(Gugging Swallowing Screen)

1. Preliminary Investigation /Indirect Swallowing Test

	YES	*NO*
Vigilance *(The patient must be alert for at least for 15 minutes)*	1 ☐	0 ☐
Cough and/or throat clearing *(voluntary cough)* *(Patient should cough or clear his or her throat twice)*	1 ☐	0 ☐
Saliva Swallow: • Swallowing successful	1 ☐	0 ☐
• Drooling	0 ☐	1 ☐
• Voice change (hoarse, gurgly, coated, weak)	0 ☐	1 ☐
SUM:		(5)
	1 - 4= Investigate further[1] 5= Continue with part 2	

2. Direct Swallowing Test (Material: Aqua bi, flat teaspoon, food thickener, bread)

In the following order:	1 →	2 →	3 →
	SEMISOLID*	LIQUID**	SOLID ***
DEGLUTITION:			
▪ Swallowing not possible	0 ☐	0 ☐	0 ☐
▪ Swallowing delayed (> 2 sec.) (Solid textures > 10 sec.)	1 ☐	1 ☐	1 ☐
▪ Swallowing successful	2 ☐	2 ☐	2 ☐
COUGH (involuntary): *(before, during or after swallowing – until 3 minutes later)*			
▪ Yes	0 ☐	0 ☐	0 ☐
▪ No	1 ☐	1 ☐	1 ☐
DROOLING:			
▪ Yes	0 ☐	0 ☐	0 ☐
▪ No	1 ☐	1 ☐	1 ☐
VOICE CHANGE: *(listen to the voice before and after swallowing - Patient should speak „O")*			
▪ Yes	0 ☐	0 ☐	0 ☐
▪ No	1 ☐	1 ☐	1 ☐
SUM:	(5)	(5)	(5)
	1 - 4= Investigate further[1] 5= Continue Liquid	**1 - 4**= Investigate further[1] 5= Continue Solid	**1 - 4**= Investigate further[1] 5= Normal

SUM: **(Indirect Swallowing Test AND Direct Swallowing Test)** _____ (20)

*	First administer ⅓ up to a half teaspoon Aqua bi with food thickener (pudding-like consistency). If there are no symptoms apply 3 to 5 teaspoons. Assess after the 5th spoonful.
**	3, 5, 10, 20 ml Aqua bi - if there are no symptoms continue with 50 ml Aqua bi (Daniels et al. 2000; Gottlieb et al. 1996) Assess and stop the investigation when one of the criteria is observed!
***	Clinical: dry bread; FEES: dry bread which is dipped in coloured liquid
[1]	Use functional investigations such as Videofluoroscopic Evaluation of Swallowing (VFES) , Fiberoptic Endoscopic Evalutation of Swallowing (FEES)

(三)反覆唾液吞嚥測試

讓病人直立端坐，30 秒內盡可能連續進行吞嚥動作，愈多愈好，施測者觸摸喉結處，感知喉頭上抬的吞嚥動作。若喉頭上抬的吞嚥次數在 30 秒內小於 3 次即判定為有吞嚥困難的高風險。

反覆唾液吞嚥測試(Repetitive Saliva Swallowing Test, RSST)的優點是安全、容易執行、省時，且相對客觀，在過去研究中發現此測驗可預測中風病人的吸入危險性。然而反覆唾液吞嚥測試僅測量簡易單一指標，缺乏對吞嚥功能其他面向的瞭解。

(四)改良式喝水試驗

以 3 ml 冷水注入病人的口腔底部進行吞嚥，觀察吞嚥過程症狀，評分為 1~5 分，1 分最嚴重，4 分表示無嗆到反應或濕泡聲，若連續 2 次無異常吞嚥則評為 5 分（表 3-5）。**改良式喝水試驗**(Modified Water Swallowing Test, MWST)容易執行且省時，但於中風病人偵測吸入情形的敏感度較低僅約 50%，特異性則約 80%。

表 3-5　改良式喝水試驗

評分	狀態
1 分	沒有吞嚥，有嗆咳和／或呼吸變化
2 分	有吞嚥，有呼吸變化
3 分	有吞嚥，呼吸良好，有嗆咳和／或濕囉音
4 分	有吞嚥，沒有嗆咳，沒有嗆咳和／或濕囉音
5 分	判定為 4 分的情況，且能在 30 秒內繼續進行 2 次吞嚥動作
無法判定	沒有吞嚥、吐出，或沒有反應

CVA (Ischemic/Hemorrhage)

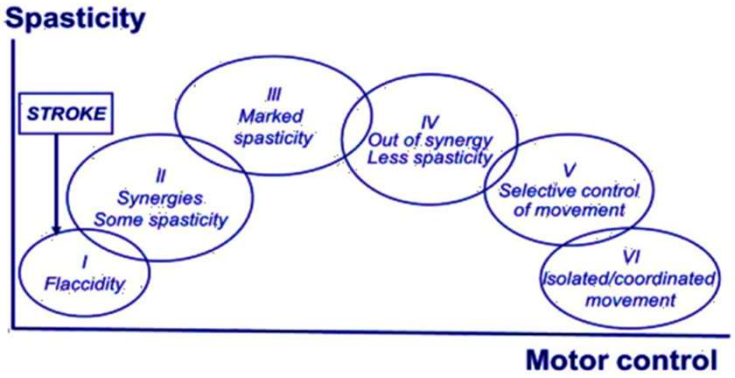
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UE(P):	I	NP	UE(P):	0	5
UE(D):	I	NP	UE(D):	0	5
LE:	I	NP	LE(P):	0	5
			LE(D):	0	5

ᵏModified Ashworth Scale	R	L
upper limb:	0	0
lower limb:	0	0

ᵏUrine / stool: incontinence / continence
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rolling: R2L poor, sitting: poor, standing: unable, walking: unable
ᵏActivities of daily life: totally dependent

Brunnstrom stage

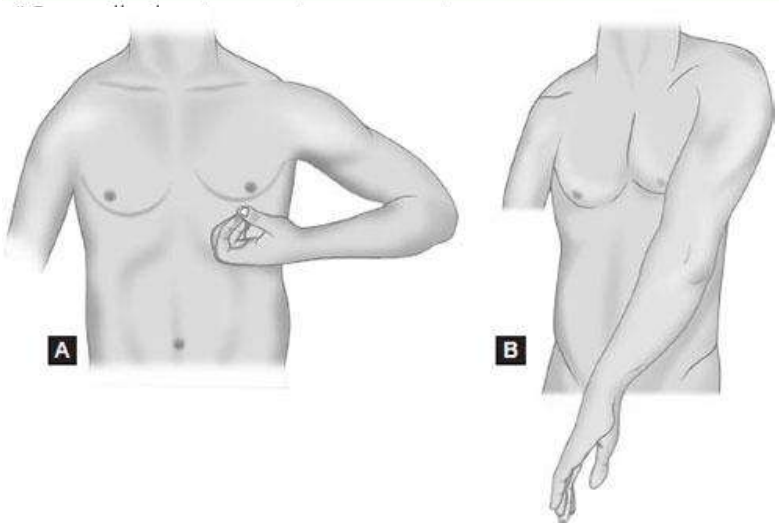


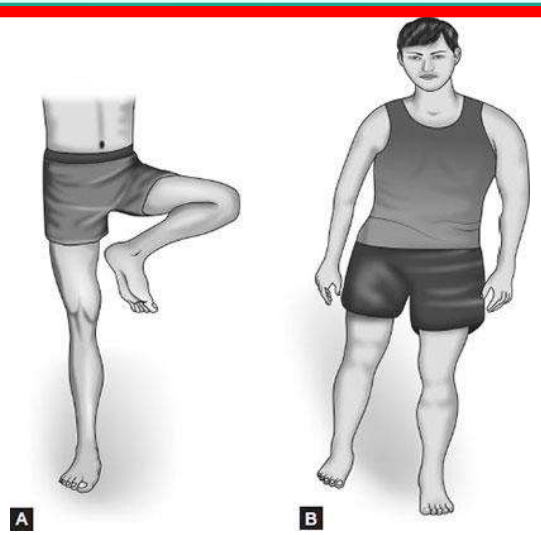
第一期：患肢軟弱無力**(flaccid)，無自主動作** (DTR -, MP = 0)
第二期：仍無明顯自主動作，但可能會有聯合運動(associated movement)，**DTR及spasticity開始增強** (DTR +, MP = 0)
第三期：**開始出現自主的動作**，但只有**協同動作(synergy movement)**，病人DTR 和spasticity 達到最強 (DTR ++, MP > 0)
第四期：協同動作開始消失，**有關節獨立動作**
第五期：比第四期更進步，可以選擇性移動大部分的關節
第六期：所有單獨關節動作都可以做，肌肉張力也接近正常，只有在速度快或做交替動作時才會出現異常
N/NP：正常肢體

Synergy Pattern

Table 15.10 Obligatory Synergy Patterns Following Stroke

	Flexion Synergy Components	Extension Synergy Components
Upper extremity	Scapular retraction/elevation or hyperextension Shoulder abduction, external rotation Elbow flexion* Forearm supination Wrist and finger Flexion	Scapular protraction Shoulder adduction,* internal rotation Elbow extension Forearm pronation* Wrist and finger flexion
Lower extremity	Hip flexion,* abduction, external rotation Knee flexion Ankle dorsiflexion, inversion Toe dorsiflexion	Hip extension, adduction,* internal rotation Knee extension* Ankle plantarflexion,* inversion Toe plantarflexion

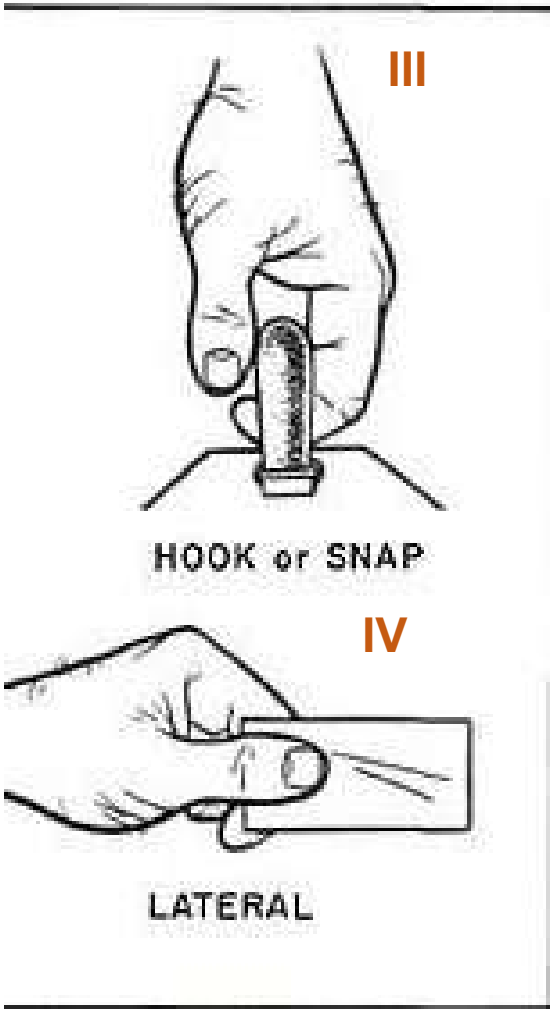


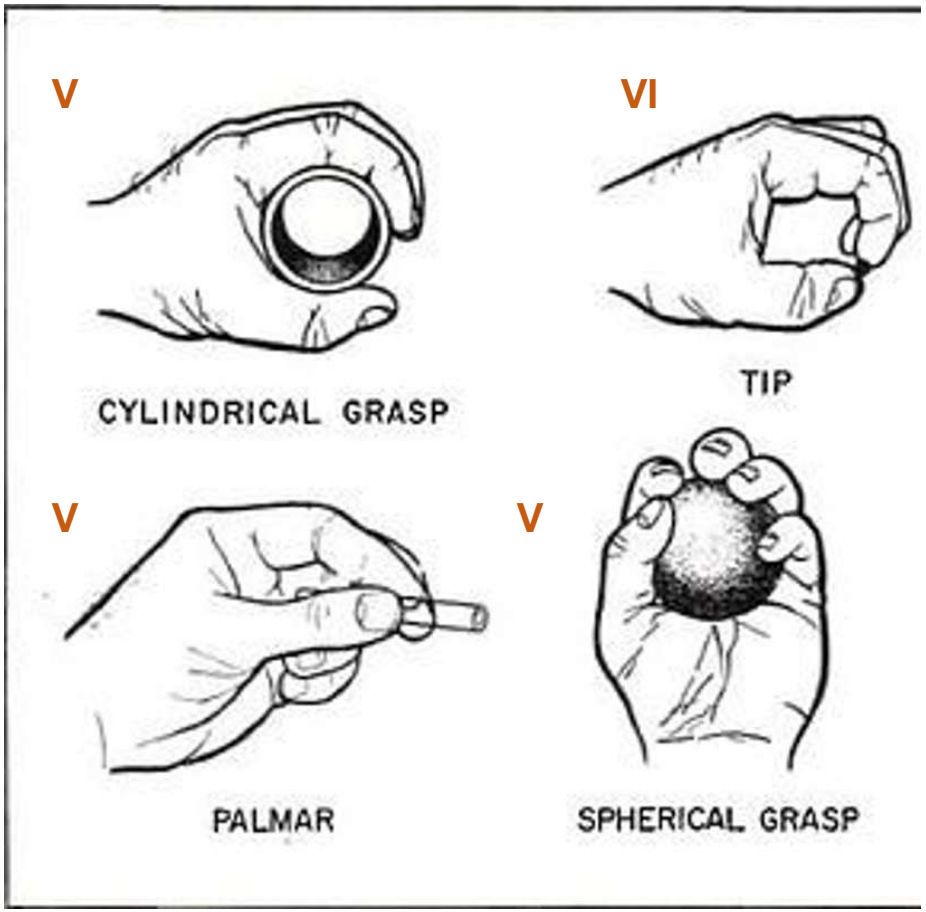


	測驗動作一	測驗動作二	測驗動作三
階層 1	上肢癱軟，無任何動作		
階層 2	平常無法自行做出動作,但打哈欠或嘔噁時，手臂自己彎起來或伸直	或是健側手出力，例：與其他人比腕力時，患側手肘跟著彎曲	或是站立或走路時,手肘不自覺彎曲
階層 3	可以主動做出動作,但無法控制動作方式	欲將手臂向前伸直平舉,手肘卻只能彎著 	欲將手肘伸直時,手臂會向對側下壓 
階層 4	可做出手臂伸直向前平舉 	可做出稍息動作(手碰腰後) 	手肘彎曲 90 度，可將手心上下翻轉  
階層 5	可將手肘伸直，向上舉高至 180 度 	可做出向側邊伸直平舉動作 	可將手臂向前平舉，手心上下翻轉  
階層 6	患側肩、肘關節可做出大部分自己想做的動作，但可能有動作遲鈍現象		

Brunnstrom Stage 近端/遠端

- 上肢近端
  - IV: 手臂伸直向前平舉、手背背後、手肘**彎曲**可 supination/pronation
  - V: 手臂可舉側邊水平或高舉過頭、手肘**伸直**可 supination/pronation
- 上肢遠端
  - III: 可握拳、但打不開
  - IV: lateral prehension, **finger semi-extension**
  - V:比 123, palmar prehension, finger full extension


III
HOOK or SNAP
IV
LATERAL


V
CYLINDRICAL GRASP
VI
TIP
V
PALMAR
V
SPHERICAL GRASP

Brunnstrom Stage 下肢

I: 癱軟

II: DTR(+), associated movement / reaction

III: 自主動作, Synergy Pattern

IV: 坐著打拍子

V: 站著打拍子

VI: 近乎正常

MRC Muscle power scale

Score	Description
0	No contraction
1	Flicker or trace of contraction
2	Active movement, with gravity eliminated
3	Active movement against gravity
4	Active movement against gravity and resistance
5	Normal power

<Rehabilitation assessment>
*Consciousness: mildly stupor, Glasgow coma scale: E4VaM6
*O2: room air
*Intubation: (+) NG, (-) Foley cath (remove today), (-) tracheostomy
*Swallowing: dysphagia, drooling(-), choking(+)
*Speech: aphasia and dysarthria
 naming: poor, 筆(O), 枕頭x
 comprehension: fair, poor , equivocal 1-step order
 fluency: poor, 1-10: x
 repetition: poor-fair, 5 words

*Brunnstrom stage:	R	L	Muscle power:	R	L
UE(P):	I	NP	UE(P):	0	5
UE(D):	I	NP	UE(D):	0	5
LE:	I	NP	LE(P):	0	5
			LE(D):	0	5

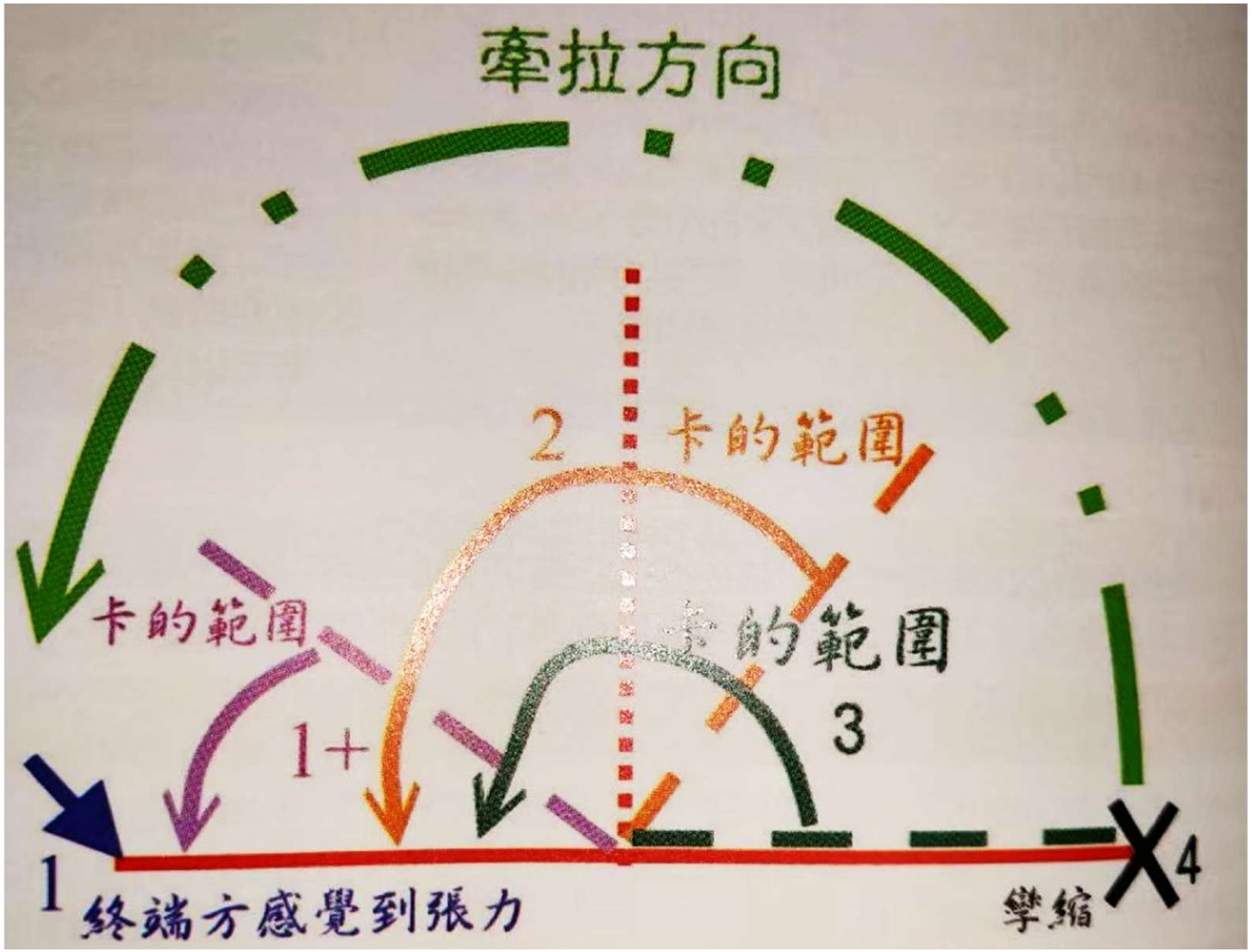
*Modified Ashworth Scale	R	L
upper limb:	0	0
lower limb:	0	0

*Urine / stool: incontinence / continence
*Functional status:(unable, poor, fair, good)
 rolling: R2L poor, sitting: poor, standing: unable, walking: unable
*Activities of daily life: totally dependent

Spasticity

- **Velocity-dependent** increase in muscle tone
- Upper motor neuron sign

Modified Ashworth Scale



<Rehabilitation assessment>
*Consciousness: mildly stupor, Glasgow coma scale: E4VaM6
*O2: room air
*Intubation: (+) NG, (-) Foley cath (remove today), (-) tracheostomy
*Swallowing: dysphagia, drooling(-), choking(+)
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 comprehension: fair, poor , equivocal 1-step order
 fluency: poor, 1-10: x
 repetition: poor-fair, 5 words

*Brunnstrom stage:	R	L
UE(P):	I	NP
UE(D):	I	NP
LE:	I	NP

Muscle power:	R	L
UE(P):	0	5
UE(D):	0	5
LE(P):	0	5
LE(D):	0	5

*Modified Ashworth Scale	R	L
upper limb:	0	0
lower limb:	0	0

*Urine / stool: incontinence / continence
*Functional status:(unable, poor, fair, good)
 rolling: R2L poor, sitting: poor, standing: unable, walking: unable
*Activities of daily life: totally dependent

改良Rankin量表(Modified Rankin Scale, MRS)

中風殘疾患者評估

患 者 狀 況	評分標準
完全無症狀	0
儘管有症狀，但無明顯功能障礙，能完成所有日常工作和生活	1
輕度殘疾，不能完成病前所有活動，但不需幫助能照料自己的日常事務	2
中度殘疾，需部分説明，但能獨立行走	3
中重度殘疾，不能獨立行走，日常生活需別人幫助	4
重度殘疾，臥床，二便失禁，日常生活完全依賴他人	5

改良Rankin量表(Modified Rankin Scale, MRS)

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患 者 狀 況	評分標準
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輕度殘疾，不能完成病前所有活動，但不需幫助能照料自己的日常事務	2
中度殘疾，需部分説明，但能獨立行走	3
中重度殘疾，不能獨立行走，日常生活需別人幫助	4
重度殘疾，臥床，二便失禁，日常生活完全依賴他人	5

TBI

Traumatic Brain Injury

```
<Rehabilitation assessment>
*Consciousness: E4V4-5M6, confused obey
*Rancho Los Amigos scale: VI
*Intubation: (-) NG, (-) Foley cath, (-) tracheostomy
*Swallowing: no dysphagia
*Speech: impaired cognition due to TBI
 naming: poor-fair(筆O, 戒指X, 手錶O), comprehension: poor-fair, confused obey, fluency: fair, repetition: fair
*Brunnstrom stage:  R    L    Muscle power:  R    L
           UE(P):  V    V          UE(P):  4+   4
           UE(D):  V    V          UE(D):  4+   4
              LE:  V    V          LE(P):  4    4
                                   LE(D):  4    4
*Modified Ashworth Scale   R   L
            upper limb:    0   0
            lower limb:    0   0
*Urine / stool: continence / continence
*Functional status:(unable, poor, fair, good)
 rolling: fair, sitting: fair, standing: poor-fair, walking: poor-fair, need assistence
*Activities of daily life: partially dependent
```

TBI

RANCHOS LOS AMIGOS SCALE (Cognition recovery for TBI)

Level	
I	No Response: Total Assistance
II	Generalized Response: Total Assistance
III	Localized Response: Total Assistance
IV	Confused, Agitated: Maximal Assistance
V	Confused, Inappropriate Non-Agitated: Maximal Assistance
VI	Confused, Appropriate: Moderate Assistance
VII	Automatic, Appropriate: Minimal Assistance for Daily Living Skills
VIII	Purposeful, Appropriate: Stand-By Assistance
IX	Purposeful, Appropriate: Stand-By Assistance on Request
X	Purposeful, Appropriate: Modified Independent

低認知程度	
I.No Response(無反應)	病人出現熟睡的狀態，且**對任何刺激沒有反應**。
II.Generalized Response (一般反應)	對所有的刺激都會產生**全身性反應**，反應皆相同，反應有可能是因生理的改變、明顯身體的活動，通常會限制或延遲的發出聲音，**是從痛最先開始有反應的**。
III.Localized Response (局部反應)	1.對特定的刺激會有特定的反應，反應和立即刺激的型式有關聯，**手腳的縮回或是出聲，對於疼痛有立即的反應**，簡單的命令像是閉上眼睛，手用力會延遲的樣子。 2.病人有時會表現出對自身感到不安，藉由**拉鼻胃管、導管或抵抗束縛**的表現，也許會表現出傾向親近人的反應，如果表面的刺激被移除了就會安穩的休息。

TBI

中低認知程度

IV.Confused-Agitated (混淆－躁動)	1.病人處於活動增加的狀態及興奮和精神混亂的情形，行為古怪，無計劃性的和環境互動，對於一些刺激可能會大叫，或是發出刺耳的叫聲。 2.會有攻擊性的行為，企圖想移除束縛及管子，或是爬出床，對環境注意明顯的短暫，選擇性的注意通常是不存在的。 3.**缺少任何回想的能力**，對於外來資訊處理能力嚴重的下降。 4.無法辨識人或是事物，無法做合作性的活動，沒有較大的協助無法完成自我照顧，可能有困難的去完成動作，像是去坐、拿的活動

TBI

中高認知程度

V.Confuse-Inappropriate (混淆－不適應)	1. 病人可對簡單一致的指令有反應，然而隨著增加複雜的指令，反應會是無目的、混亂的、不完整的。 2. 病人可有顯著的注意力在環境上，但容易分心及缺乏集中注意力在特定工作上的能力。 3. 語言表現經常是不適當的，在記憶中以幻想替代事實，記憶嚴重缺損不能學習新事物，對自我、身體、家人數目反應最好，對於不安或不愉快的刺激會有表現激動的行為。 4. 通常能完成自我照額，但需協助。 (不再焦躁不安，對簡單的指令有反應)
VI.Confuse-Appropriate (混淆－適應)	1. 病人表現出有目的的行為，但依賴在外來的刺激或指導，有適當的不安反應及能容許不偷快的刺激 2. 遵從簡單直接一致性和再學習新的學習任務(自我照顧)。 3. 但有關記憶方面的問題反應，可能是不適當的，對過去的記憶表現的比新的記憶更深入及詳細。 4. 對有選擇性注意力的任務受損，對自我、家人、和基本需求有更進一步的認識。(病人的行為能力更好，有好的回饋會有適當活動或有目的的動作產生)

高認知程度

VII.Automatic-Appropriate (自主－適應)	1.病人出現適當和適合的情況反應在家中和醫院內，有較淺的回想活動，對自我、家人、身體、食物、人群及和環境的互動增加。 2.但缺乏理解力，及抽象問題的解決能力，缺乏現實的主意和計劃及未來的表現， 3.能加入社會及娛樂的活動。(對自己整天生活流程有比較了解，可回想以前的事)。
VIII. **Purposeful-Appropriate** (有目的－適應)	1.病人有回想能力，可合併過去經驗及最近的事情和環境產生適當行為。 2.對於新的學習和需要可不須監督，理解力推理、忍受力、抽象思考能力都增加。(可將過去的經驗用在現在的生活裡)

SCI

SCI的評估方式
ASIA Impairment Scale (AIS)

The Journal of Spinal Cord Medicine

Taylor & Francis
Taylor & Francis Group

J Spinal Cord Med. 2011 Nov; 34(6): 535–546.
doi: 10.1179/204577211X13207446293695

PMCID: PMC3232636
PMID: 22330108

International standards for neurological classification of spinal cord injury (Revised 2011)

Steven C. Kirshblum,[1,2] Stephen P. Burns,[3] Fin Biering-Sorensen,[4] William Donovan,[5] Daniel E. Graves,[6] Amitabh Jha,[7] Mark Johansen,[7] Linda Jones,[8] Andrei Krassioukov,[9] MJ Mulcahey,[10] Mary Schmidt-Read,[11] and William Waring[12]

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J Spinal Cord Med

SCI

感覺怎麼測？

Pin prick

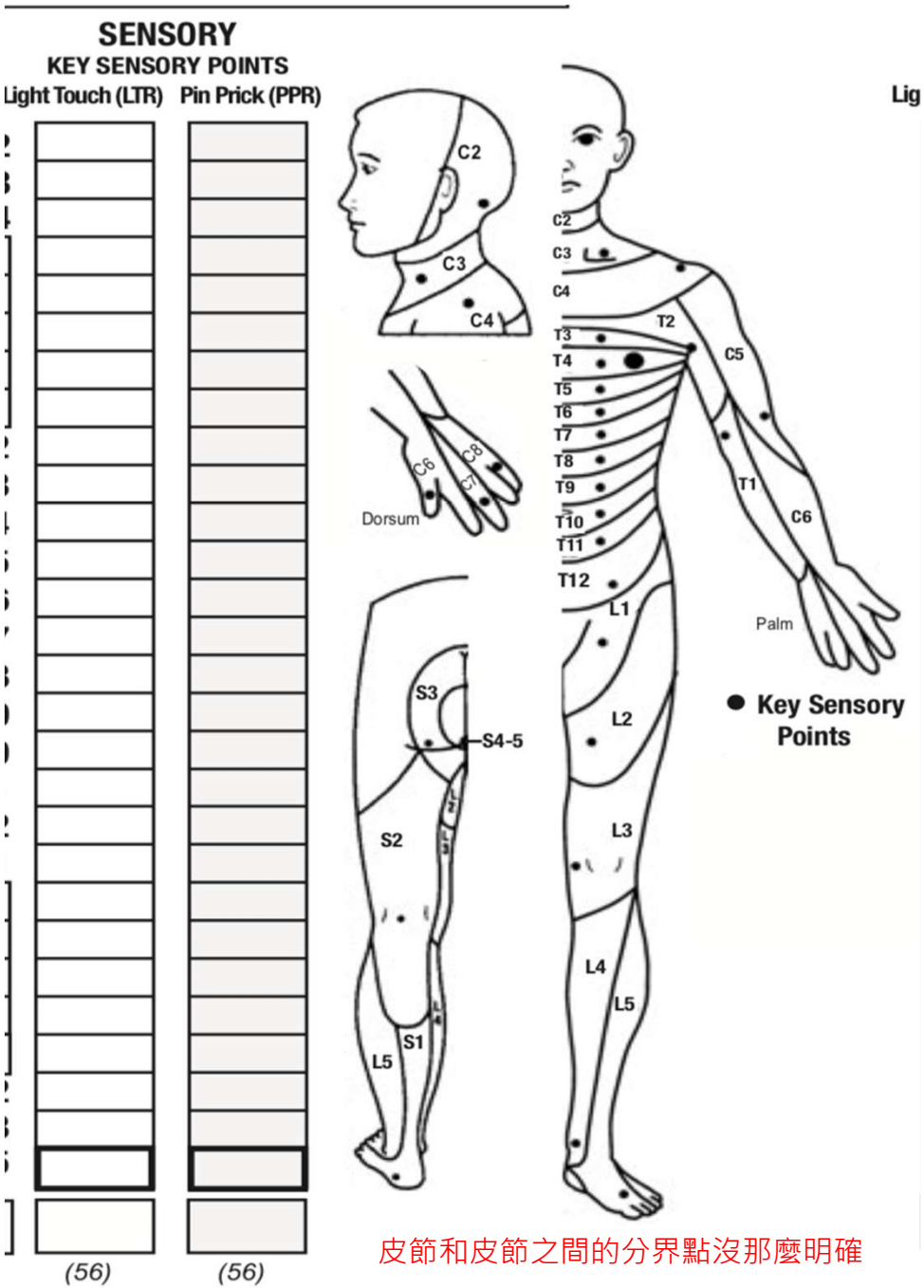
- 定義是要能分辨尖鈍(sharp/dull discrimination)
- 使用可拋棄式的安全針頭，一端為尖，一端為鈍(sharp/dull)
- 每個key point均要和臉部進行比較
- 0分：病人沒有能力分辨是尖端還是鈍端
- 1分：病人可以分辨尖端或鈍端，但是感覺跟臉不一樣(hyperesthesia也算)
- 2分：病人可以分辨尖端或鈍端，並且感覺跟臉一樣

Light touch

- 使用一縷棉花(a tapered wisp of cotton)，從距離皮膚1公分以內，輕輕碰觸皮膚一下。
- 病人的眼睛需閉上。
- 0分：沒有感覺
- 1分：感覺和臉不一樣(hyperesthesia也算)
- 2分：感覺和臉一樣

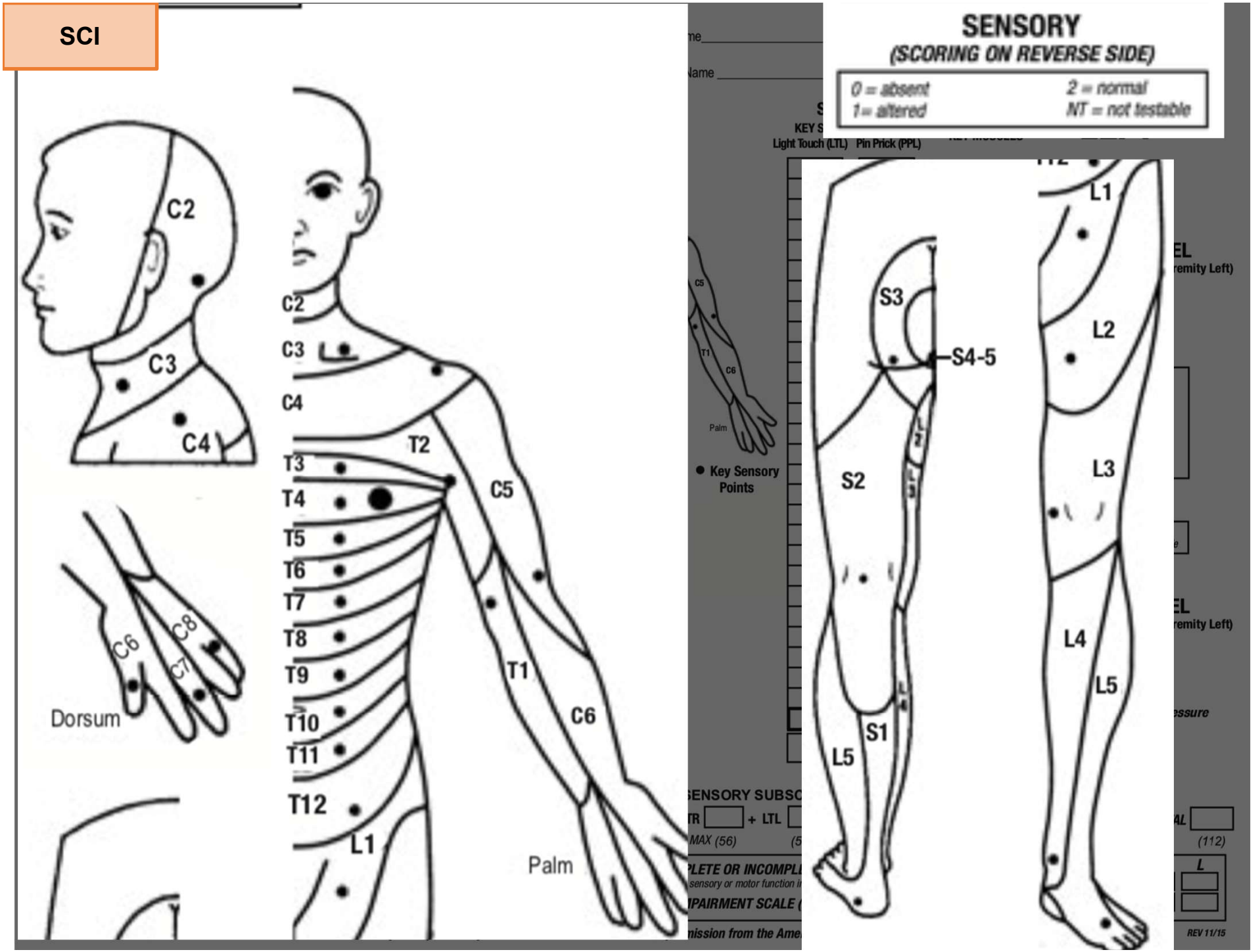
雙側Pin prick與light touch均為2分，該節段才算正常。

SCI



皮節和皮節之間的分界點沒那麼明確

SCI


SENSORY
(SCORING ON REVERSE SIDE)
0 = absent
1 = altered
2 = normal
NT = not testable
Light Touch (LTL)
Pin Prick (PPL)
C2
C3
C4
C5
C6
C7
C8
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
L1
L2
L3
L4
L5
S1
S2
S3
S4-5
Palm
Dorsum
Key Sensory Points

SCI

Spinal Cord Injury

<<Rehabilitation assessment>>
*Consciousness: agitated, Glasgow coma scale:E4V4M6
*Pressure sore: nil
*Intubation: NG (+), Foley cath (+), CVP (+)

10 key muscles

*Muscle power:		R	L	R	L	
Elbow flexors	(C5)	5	5	3	0	(L2)Hip flexors
Wrist extensors	(C6)	5	5	3	0	(L3)Knee extensors
Elbow extensors	(C7)	5	5	3	0	(L4)Ankle dorsiflexors
Finger flexors	(C8)	5	5	3	0	(L5)Long toe extensors
Finger abductors	(T1)	5	5	3	0	(S1)Ankle plantar flexors

*Sensory level: around T6
*Motor level: above Below T1
*Neurological level: suspect T6
*Anal tone (+), Bulbocavernous reflex (+)

*Modified Ashworth Scale	R	L
upper limb:	0	0
lower limb:	0	1

*Urine / stool: Foley / incontinence
*Functional status:(unable, poor, fair, good)
rolling: poor, sitting: unalbe, standing: unable, walking:unable
*Activities of daily life: partially dependent

SCI

INTERNATIONAL STANDARDS FOR NEUROLOGICAL CLASSIFICATION OF SPINAL CORD INJURY (ISNCSCI)

ISCOS — International Spinal Cord Society

Patient Name _____ Date/Time of Exam _____

Examiner Name _____ Signature _____

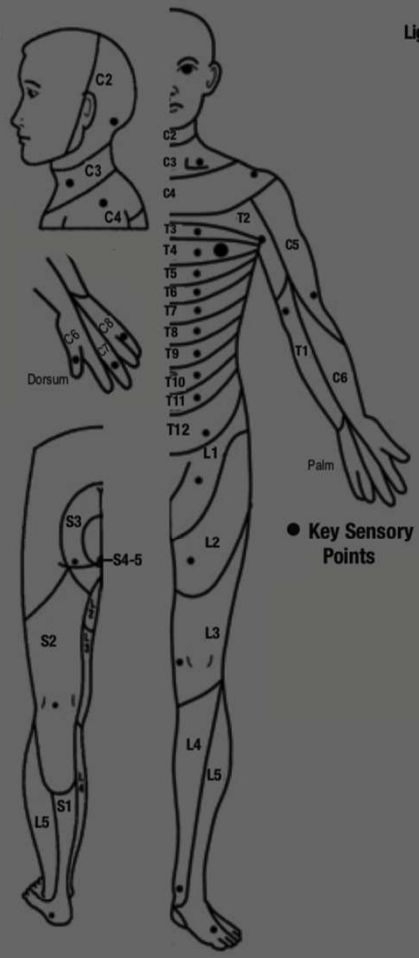
RIGHT

10 key muscles

MOTOR KEY MUSCLES		SENSORY KEY SENSORY POINTS Light Touch (LTR)	Pin Prick (PPR)
		C2	
		C3	
		C4	
UER (Upper Extremity Right) — Elbow flexors	C5		
Wrist extensors	C6		
Elbow extensors	C7		
Finger flexors	C8		
Finger abductors (little finger)	T1		
		T2	
		T3	
		T4	
		T5	
		T6	
		T7	
		T8	
		T9	
		T10	
		T11	
		T12	
LER (Lower Extremity Right) — Hip flexors	L2	L1	
Knee extensors	L3		
Ankle dorsiflexors	L4		
Long toe extensors	L5		
Ankle plantar flexors	S1		
		S2	
		S3	
		S4-5	
		RIGHT TOTALS (MAXIMUM) (50)	(56) (56)

Comments (Non-key Muscle? Reason for NT? Pain?):

(VAC) Voluntary Anal Contraction (Yes/No) ☐



● Key Sensory Points

LEFT

SENSORY KEY SENSORY POINTS Light Touch (LTL)	Pin Prick (PPL)	MOTOR KEY MUSCLES	
		C2	
		C3	
		C4	
		C5	Elbow flexors
		C6	Wrist extensors
		C7	Elbow extensors
		C8	Finger flexors
		T1	Finger abductors (little finger)
		T2	
		T3	
		T4	
		T5	
		T6	
		T7	
		T8	
		T9	
		T10	
		T11	
		T12	
		L1	
		L2	Hip flexors
		L3	Knee extensors
		L4	Ankle dorsiflexors
		L5	Long toe extensors
		S1	Ankle plantar flexors
		S2	
		S3	
		S4-5	
(56)	(56)	LEFT TOTALS (MAXIMUM) (50)	

UEL (Upper Extremity Left)

LEL (Lower Extremity Left)

MOTOR (SCORING ON REVERSE SIDE)

0 = total paralysis
1 = palpable or visible contraction
2 = active movement, gravity eliminated
3 = active movement, against gravity
4 = active movement, against some resistance
5 = active movement, against full resistance
5* = normal corrected for pain/disuse
NT = not testable

SENSORY (SCORING ON REVERSE SIDE)

0 = absent 2 = normal
1 = altered NT = not testable

(DAP) Deep Anal Pressure (Yes/No) ☐

MOTOR SUBSCORES

UER ☐ + UEL ☐ = UEMS TOTAL ☐ (MAX (25) (25) (50))

LER ☐ + LEL ☐ = LEMS TOTAL ☐ (MAX (25) (25) (50))

SENSORY SUBSCORES

LTR ☐ + LTL ☐ = LT TOTAL ☐ (MAX (56) (56) (112))

PPR ☐ + PPL ☐ = PP TOTAL ☐ (MAX (56) (56) (112))

NEUROLOGICAL LEVELS (Steps 1-5 for classification as on reverse)

	R	L
1. SENSORY	☐	☐
2. MOTOR	☐	☐

3. NEUROLOGICAL LEVEL OF INJURY (NLI) ☐

4. COMPLETE OR INCOMPLETE? ☐ (Incomplete = Any sensory or motor function in S4-5)

5. ASIA IMPAIRMENT SCALE (AIS) ☐

(In complete injuries only) ZONE OF PARTIAL PRESERVATION (Most caudal level with any innervation)

	R	L
SENSORY	☐	☐
MOTOR	☐	☐

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REV 11/15

SCI

UER (Upper Extremity Right)	*Elbow flexors*	C5	
	Wrist extensors	C6	
	Elbow extensors	C7	
	Finger flexors	C8	
	Finger abductors (little finger)	T1	

LER (Lower Extremity Right)	*Hip flexors*	L2	
	Knee extensors	L3	
	Ankle dorsiflexors	L4	
	Long toe extensors	L5	
	Ankle plantar flexors	S1	

MOTOR
(SCORING ON REVERSE SIDE)

0 = total paralysis
1 = palpable or visible contraction
2 = active movement, gravity eliminated
3 = active movement, against gravity
4 = active movement, against some resistance
5 = active movement, against full resistance
5 = normal corrected for pain/disuse*
NT = not testable

SCI

Muscle Strength怎麼測？

0	total paralysis
1	palpable or visible contraction
2	active movement, full range of motion (ROM) with gravity eliminated
3	active movement, full ROM against gravity
4	active movement, full ROM against gravity and moderate resistance in a muscle specific position
5	(normal) active movement, full ROM against gravity and full resistance in a muscle specific position expected from an otherwise unimpaired person
5*	(normal) active movement, full ROM against gravity and sufficient resistance to be considered normal if identified inhibiting factors (i.e. pain, disuse) were not present
NT	not testable (i.e. due to immobilization, severe pain such that the patient cannot be graded, amputation of limb, or contracture of >50% of the range of motion).

SCI

Steps in Classification

The following order is recommended for determining the classification of individuals with SCI.

1. Determine sensory levels for right and left sides.

The sensory level is the most caudal, intact dermatome for both pin prick and light touch sensation.

先測 sensory level
該節段Pinprick與light touch皆為2分 (正常)

2. Determine motor levels for right and left sides.

Defined by the lowest key muscle function that has a grade of at least 3 (on supine testing), providing the key muscle functions represented by segments above that level are judged to be intact (graded as a 5)

Note: in regions where there is no myotome to test, the motor level is presumed to be the same as the sensory level, if testable motor function above that level is also norm

再來定 motor level
該節段MMT ≧3分，且上一節必須為5分

3. Determine the neurological level of injury (NLI)

This refers to the most caudal segment of the cord with intact sensation and antigravity (3 or more) muscle function strength, provided that there is normal (intact) sensory and motor function rostrally respectively.

The NLI is the most cephalad of the sensory and motor levels determined in steps 1 and 2.

感覺與運動各自最低節段，取較高者決定NLI
若無運動節段(T-spine)，則以感覺節段定level

SCI

4. Determine whether the injury is Complete or Incomplete.

(i.e. absence or presence of sacral sparing)

*If voluntary anal contraction = **No** AND all S4-5 sensory scores = **0***
*AND deep anal pressure = **No,** then injury is **Complete.***
*Otherwise, injury is **Incomplete.***

最終確定損傷是 complete or incomplete
沒有VAC + 沒有DAP + 雙側S4/5 dermatome = 0: complete (AIS = A)
上述三項任一不符合: incomplete (開始分AIS B、C、D)

Voluntary Anal Contraction (VAC) (肛門外括約肌，陰部神經，S2-4)
- 將手指放入病人肛門，請病人「收縮肛門夾我的手指並hold住」
- YES → Motor incomplete injury (至少AIS=C)

Deep Anal Pressure (DAP)
- 將手指放入病人肛門，問病人是否有感覺
- 屬於S4、S5 Dermatome

SCI

5. Determine ASIA Impairment Scale (AIS) Grade:

Is injury Complete? If YES, AIS=A and can record ZPP (lowest dermatome or myotome on each side with some preservation)

沒有VAC + 沒有DAP = AIS A

NO ↓

Is injury Motor Complete? If YES, AIS=B

沒有VAC + motor level三節以下MMT=0, 為AIS B

NO ↓

(No=voluntary anal contraction OR motor function more than three levels below the motor level on a given side, if the patient has sensory incomplete classification)

Are at least half (half or more) of the key muscles below the neurological level of injury graded 3 or better?

NO ↓ AIS=C

YES ↓ AIS=D

如果非AIS A or B, 接著看NLI以下的Key muscle
- 超過一半≧3, AIS D
- if not, AIS C

If sensation and motor function is normal in all segments, AIS=E

Note: AIS E is used in follow-up testing when an individual with a documented SCI has recovered normal function. If at initial testing no deficits are found, the individual is neurologically intact; the ASIA Impairment Scale does not apply.

SCI

Sensory level

PP	R	L
C2	2	2
C3	2	2
C4	2	2
C5	2	2
C6	2	1
C7	0	0
C8	0	0
C9	0	0

LT	R	L
C2	2	2
C3	2	2
C4	2	2
C5	2	2
C6	2	2
C7	0	1
C8	0	0
C9	0	0

Sensory level: C5

SCI

Motor level

	R	L
C5	5	5
C6	5	5
C7	4	3
C8	0	0
T1	0	0

Motor level: C7

SCI

Neurogical level

PP	R	L
C2	2	2
C3	2	2
C4	2	2
C5	2	2
C6	2	1
C7	0	0
C8	0	0
C9	0	0

LT	R	L
C2	2	2
C3	2	2
C4	2	2
C5	2	2
C6	2	2
C7	0	1
C8	0	0
C9	0	0

	R	L
C5	5	5
C6	5	5
C7	4	3
C8	0	0
T1	0	0

Sensory level: C5

Motor level: C7

Neurological level: C5

延伸閱讀

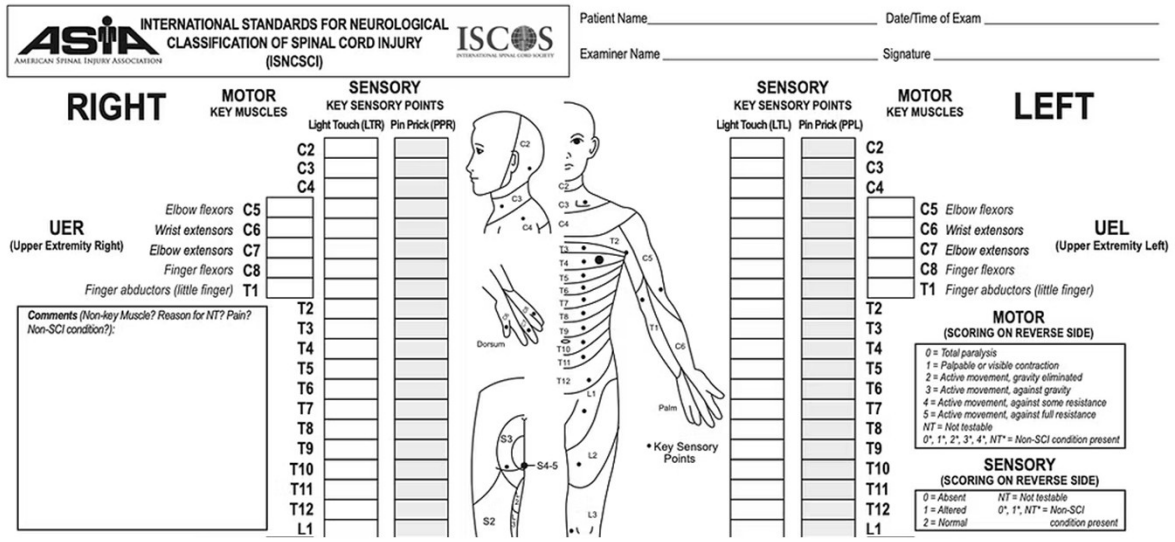
https://www.drmbesuperior.com/post/%E8%84%8A%E9%AB%93%E6%90%8D%E5%82%B7%E8%A9%95%E4%BC%B0%E5%B7%A5%E5%85%B7-isncsci-asia-score

https://reurl.cc/3yoKkl

Dr. M · Sep 16, 2020 · 12 min read

脊髓損傷評估工具 ISNCSCI ASIA Score

Updated: Jun 17, 2023



不適用ASIA Impairment Scale的clinical syndromes

- Central Cord Syndrome
- Brown-Sequard Syndrome
- Anterior Cord Syndrome
- Posterior Cord Syndrome
- Conus Medullaris Syndrome
- Cauda Equina Syndrome

Thank you for your attention